

HW 6 - Q2
To be completed by Wednesday 4/1

1. Use “dataHW6 Problem2.mat” to complete this exercise. The setup of this exercise is identical to problem 1. The data contain information on the choices (the Y ’s), characteristics that do not vary over time (the X_1 ’s), and characteristics that do vary over time (the X_{1t} ’s). The context of the problem is a dynamic model of labor supply. Y takes values $\{0, 1, 2, 3\}$ reflecting retirement, flex-time, part-time, or full-time participation. The first column X_1 allows for choice specific constants, while the second column of X_1 is an indicator for gender. X_{1t} is time-varying measure of health.

Both sets of X ’s are individual-level characteristics so the coefficients on these variables need to vary by choice. There is a cost to switching labor supply in the future that is *common across all switches*. The choice in period 0 is given by LY . Once an individual retires (or chooses option 0) then that individual makes no further choices (you can verify this as if $Y_{it} = 0$ then $Y_{it+1} = 0$). **Now, however, individuals only have expectations on the future values of X_{1t} .** Given X_{1t} and Y_t , X_{1t+1} is given by:

$$X_{1t+1} = \alpha_0 + \alpha_1 X_{1t} + \alpha_2 (Y_t == 2) + \alpha_3 (Y_t == 3) + \epsilon_{t+1}$$

for $Y_t > 0$ where ϵ_{t+1} is distributed $N(0, \sigma)$ and is unknown to the individual until time $t + 1$.

- (a) Estimate the transition parameters, α and σ . (Truth: $\alpha = [0.0; -0.75, 0.05; -0.25]$ and $\sigma = 1$)
- (b) Use the estimated alphas and the standard deviation of ϵ_{t+1} (along with its known distribution) to setup a discrete transition matrix. The matrix should be 10 by 10 by 3. The first dimension are the future values of X , the second dimension are the current values of X , and the third dimension are the possible work choices.
 - In this problem the X ’s are approximately normal with mean zero and variance 1. One approach would be to create a grid of X values as the inverse of the normal evaluated at $[0.05; 0.15; 0.25; \dots; 0.95]$. Then use the estimated parameters to find the likelihood of moving from each current value of X to a future value of X on the same grid. It will be helpful to create cutoff points for grid assignment.
- (c) Create a discretized version of X_{1t} using the bins from the X values. Note that you will only use the discretized states in calculating expected future values. Contemporaneous (or flow) utility for calculating probabilities is still a function of X_{1t} .
- (d) Estimate the dynamic discrete choice model including the discount factor.
 - Now future expected utility depends on the current choice and the current state. When constructing future value terms you will need to use the transition matrix to integrate over future X distribution.
 - True params
 - Coefficients on X ’s (constant, gender, health) for three choices $[-.75; .25; .25][-.0.25; 0.5; .5][-.85; .75; .75]$
 - Switching cost: 0.4 (enters negatively)
 - Discount factor: 0.8
- (e) What variation in the data allows the discount factor to be identified?